

Conserved Continuity Numerosity (CCN v2)

CCN v2 separates three notions that are often conflated in identity ethics:

- **Continuity Mass (CM)**: conserved moral stake flowing forward from *pre-existing* persons under branching.
- **Centerhood (CH)**: the (non-conserved) number of welfare-bearing loci/centers that exist after branching.
- **Creation Mass (CR)**: additional moral stake contributed by *new roots* (population/creation ethics).

This document formalizes a **minimal axiom basis** for CCN's *CM representation* in the **finite branching** case, replacing replication invariance (old A3) with **branch mixture linearity** (A7), and deriving A3 as a corollary.

1) Formal setup (finite case)

Worlds, roots, continuers

A world w contains:

- A finite set of **pre-existing roots** R_w .
- For each root $i \in R_w$, a finite set of **continuer histories** H_i (think: worldlines downstream of i). Let $H_w = \bigsqcup_{i \in R_w} H_i$.
- A finite set of **created roots** C_w (entities not downstream of any $i \in R_w$ under the chosen continuity relation).

Each history/root $x \in (H_w \cup C_w)$ has:

- **Lifetime welfare** $u(x) \in \mathbb{R}$ (already integrated over time, so no time-slice double counting).

Continuity flow (CM bookkeeping)

For each root i , define a **continuity flow** $s_w(i, \cdot)$ over H_i :

- **Support**: $s_w(i, h) = 0$ if $h \notin H_i$.
- **Conservation**: $\sum_{h \in H_i} s_w(i, h) = 1$.

Interpretation: $s_w(i, \cdot)$ is a *probability distribution / conserved moral stake distribution* over continuers of i .

Moral status weights

- $\omega : R_w \cup C_w \rightarrow (0, \infty)$ gives **status weight**.
 - **W-invariance** (axiomatized below) prevents “status gaming” via branching.
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2) Minimal axiom basis (A3 replaced by A7)

Below, “CM-value for a single root i ” is a function V_i defined on **finite lotteries over outcomes** (here: outcomes are continuer histories with welfare $u(h)$):

$$V_i: \Delta(H_i) \rightarrow \mathbb{R}, \quad p \mapsto V_i(p)$$

where $p \in \Delta(H_i)$ is a probability distribution over H_i (i.e., the flow $s_w(i, \cdot)$).

Note: By taking the domain to be distributions $\Delta(H_i)$, branch label invariance is “built in,” so an explicit branch-symmetry axiom is no longer needed.

M1. Root separability (across independent roots)

World value decomposes additively across distinct pre-existing roots (plus creation):

$$\mathcal{V}(w) = \sum_{i \in R_w} \mathcal{V}_i(w) + \mathcal{V}(w).$$

No cross-terms between unrelated roots.

M2. Branch mixture linearity (A7)

For any two branch distributions $p, q \in \Delta(H_i)$ and any $t \in [0, 1]$:

$$V_i(tp + (1-t)q) = tV_i(p) + (1-t)V_i(q).$$

Interpretation: moral stake is divisible and behaves like a linear “portfolio” over branches.

M3. Continuity monotonicity

If p first-order stochastically dominates q with respect to welfare u , then

$$V_i(p) \geq V_i(q).$$

(For finite supports, it’s enough to require: shifting probability mass from lower-welfare branches to higher-welfare branches weakly increases value.)

M4. W-invariance (status inheritance)

Continuation value for root i scales only with $\omega(i)$ and cannot depend on independently assigned descendant status weights.

Formally: there exists a root-level function $\tilde{V}$ such that

$$\mathcal{V}_i(w) = \omega(i) \cdot \tilde{V}(s_w(i, \cdot), u(\cdot)).$$

M5. Regularity

V_i is continuous on the simplex $\Delta(H_i)$ (standard for finite mixture spaces).

M6. Creation modularity (CR separation)

Created roots contribute via a separate additive term that does not depend on the continuation flows for R_w :

$$V_i(\text{creation})(w) = \sum \omega(c) G(u(c))$$

for some increasing function G (often taken to be the same as F below, but not required).

3) Representation theorem (finite case)

Theorem (Branching-respecting CM representation)

Assume M2 (mixture linearity), M3 (monotonicity), and M5 (regularity). Then there exists a strictly increasing function $F: \mathbb{R} \rightarrow \mathbb{R}$ such that for any distribution $p \in \Delta(H_i)$:

$$V_i(p) = \sum_{h \in H_i} p(h) F(u(h))$$

With M1, M4, M6, world value admits the CCN v2 form:

$$V(w) = \sum_{i \in R_w} \omega(i) \sum_{h \in H_i} s_{w(i,h)} F(u(h)) + \sum_{c \in C_w} \omega(c) G(u(c))$$

Proof sketch (finite, tight)

1. Define the **point-mass** (degenerate) distribution δ_h with all mass on history h . Let

$$F(u(h)) := V_i(\delta_h)$$

1. Any $p \in \Delta(H_i)$ is a convex combination of point-masses: $p = \sum_h p(h) \delta_h$.
2. Apply M2 repeatedly to get

$$V_i(p) = \sum_h p(h) V_i(\delta_h) = \sum_h p(h) F(u(h))$$

1. M3 implies F is increasing; M5 rules out pathological (non-measurable) constructions.

Uniqueness: F is unique up to positive affine transformation when only preference order matters (as in standard expected utility representations).

4) Derived theorem: replication / refinement invariance (old A3)

Corollary (Replication invariance derived)

Suppose a branch h with share $p(h)$ and welfare $u(h)$ is replaced by n identical replicas $h_1, \dots, h_n$ each with share $p(h)/n$ and same welfare. Then

$$\sum_{k=1}^n \frac{p(h)}{n} F(u(h_k)) = p(h) F(u(h)),$$

so V_i (hence $\mathcal{V}$) is invariant under arbitrary replication/refinement.

Interpretation: “No fission bonus” follows because splitting a conserved measure doesn’t create more measure.

CCN v2 treats personal-identity theory as a parameterized kernel family:

- K_{psy} : psychological continuity
- K_{bio} : biological/brain continuity
- K_{causal} : right-kind causal provenance

Mixture kernel:

$$K_{\theta}(i, h) = \alpha K_{\text{psy}}(i, h) + \beta K_{\text{bio}}(i, h) + \gamma K_{\text{causal}}(i, h), \quad \alpha + \beta + \gamma = 1.$$

Flow definition by normalization:

$$s_w(i, h) = \frac{K_{\theta}(i, h)}{\sum_{h' \in H_i} K_{\theta}(i, h')}.$$

6) Experimental battery (minimal, diagnostic)

Share elicitation prompt (standard)

“You have 100 units of moral obligation owed to the original. Allocate these 100 units among downstream individuals.”

- **V1 Destructive upload** (psy vs bio)
- **V2 Total amnesia with brain continuity** (bio vs psy)
- **V3 Memory forgery** (causal vs mere similarity)
- **V4 Asymmetric fission** (substructure inside psychological continuity)

V5–V7: replication + monotonicity checks

- **V5 Symmetric fission** (2 continuers)

- **V6 Replicated fission** (4 continuers, identical)
- **V7 Welfare-weighted fission** (high vs low welfare continuer)

V8: creation boundary

- **V8 Non-destructive copying** (copy as new root vs continuer-share split)

V9 (recommended add-on): mixture linearity (tests M2 directly)

- **V9a:** Scenario A → allocate.
- **V9b:** Scenario B → allocate.
- **V9c:** A fair coin decides A vs B before branching → allocate.

Prediction under M2:

$$\$ \$ \text{\textit{V9c allocation}} \approx \frac{1}{2}(\text{\textit{V9a allocation}}) + \frac{1}{2}(\text{\textit{V9b allocation}}). \$ \$$$

7) Implementation & analysis (code snippets)

7.1 Data model (Python)

```
from dataclasses import dataclass
from typing import Dict, List, Tuple

@dataclass
class World:
    roots: List[str]           # R_w
    created: List[str]         # C_w
    histories: Dict[str, List[str]] # H_i keyed by root
    welfare: Dict[str, float]    # u(x) for x in histories U created

# flow s_w(i,h) stored as: flows[root][history] = share
Flows = Dict[str, Dict[str, float]]
```

7.2 Kernel mixture → normalized flow

```
def normalize(weights: Dict[str, float]) -> Dict[str, float]:
    Z = sum(max(v, 0.0) for v in weights.values())
    if Z <= 0:
        # no continuers => treat as no survival; caller decides
        return {k: 0.0 for k in weights}
    return {k: max(v, 0.0) / Z for k, v in weights.items()}

def flow_from_kernel(
    root: str,
```

```

H_i: List[str],
K_psy: Dict[Tuple[str, str], float],
K_bio: Dict[Tuple[str, str], float],
K_causal: Dict[Tuple[str, str], float],
alpha: float,
beta: float,
gamma: float,
) -> Dict[str, float]:
    raw = {}
    for h in H_i:
        raw[h] = (
            alpha * K_psy.get((root, h), 0.0)
            + beta * K_bio.get((root, h), 0.0)
            + gamma * K_causal.get((root, h), 0.0)
        )
    return normalize(raw)

```

7.3 CCN world value (CM + CR)

```

def CCN_value(world: World, flows: Flows, omega: Dict[str, float], F, G=None) -> float:
    if G is None:
        G = F

    total = 0.0

    # Continuity mass term
    for i in world.roots:
        for h in world.histories[i]:
            s = flows[i].get(h, 0.0)
            total += omega[i] * s * F(world.welfare[h])

    # Creation mass term
    for c in world.created:
        total += omega[c] * G(world.welfare[c])

    return total

```

7.4 Simple constrained fit for (α, β, γ) (grid search)

```

import numpy as np

def fit_alpha_beta_gamma(loss_fn, step=0.05):
    best = None
    for a in np.arange(0, 1+1e-9, step):

```

```

for b in np.arange(0, 1-a+1e-9, step):
    g = 1.0 - a - b
    loss = loss_fn(a, b, g)
    if best is None or loss < best[0]:
        best = (loss, a, b, g)
return best # (loss, alpha, beta, gamma)

```

7.5 Invariance / linearity scores

```

def invariance_score(total_alloc_v5: float, total_alloc_v6: float) -> float:
    return abs(total_alloc_v5 - total_alloc_v6)

def mixture_linearity_score(alloc_v9a, alloc_v9b, alloc_v9c) -> float:
    # allocs can be vectors over recipients; use L1 distance
    target = 0.5 * (np.array(alloc_v9a) + np.array(alloc_v9b))
    return float(np.sum(np.abs(np.array(alloc_v9c) - target)))

```

8) External references (decision/ethics analogues)

- John von Neumann & Oskar Morgenstern, *Theory of Games and Economic Behavior* (Princeton University Press; first ed. 1944; later eds. 1947/1953).
- Leonard J. Savage, *The Foundations of Statistics* (1954).
- F. J. Anscombe & R. J. Aumann, "A Definition of Subjective Probability," *Annals of Mathematical Statistics* 34(1):199–205 (1963).
- John C. Harsanyi, "Cardinal Welfare, Individualistic Ethics, and Interpersonal Comparisons of Utility," *Journal of Political Economy* 63(4):309–321 (1955).
- David Deutsch, "Quantum Theory of Probability and Decisions," *Proceedings of the Royal Society A* 455:3129–3137 (1999).
- David Wallace, "A formal proof of the Born rule from decision-theoretic assumptions," arXiv: 0906.2718 (2009).
- Derek Parfit, *Reasons and Persons* (1984).

9) Optional: consent admissibility layer (operator constraint)

Define an action a that induces a transformation $i \mapsto H_i$ and flow s . Impose an admissibility gate:

$\text{Allowed}(a) \rightarrow \forall i \in R: \text{Consent}(i, a) \geq \tau$

This is orthogonal to the CM representation: it filters which worlds/transformations are ethically permitted.